VIVEK KUMAR

Full-Stack Software Engineer • MERN Stack • Generative AI (LangChain / LCEL) • Cloud Architectures

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PROFESSIONAL SUMMARY

Associate Engineer at Virtusa and Full-Stack & Generative AI Software Engineer with proven expertise architecting high-concurrency web applications, scalable microservices, and production LLM orchestration workflows. Expert in React 18, Node.js, Express, MongoDB, and Redis Geospatial, combined with advanced implementation of LangChain LCEL chains, LangGraph multi-agent systems, semantic RAG, and real-time token streaming. Dual AWS Certified (Developer – Associate & Data Engineer – Associate).

TECHNICAL SKILLS

Frontend Architecture: React.js (v18+), Next.js, Redux Toolkit, Zustand, Tailwind CSS, WebSockets, HTML5/CSS3, Responsive UI/UX

Backend & APIs: Node.js, Express.js, RESTful APIs, GraphQL, Socket.IO, JWT Authentication, Microservices, Python (FastAPI)

Generative AI & LangChain: LangChain (LCEL Runnables), LangGraph (Stateful Multi-Agent Workflows), Semantic RAG Pipelines, ChromaDB / Pinecone / pgvector, OpenAI GPT-4o / Claude APIs, Prompt Engineering, Pydantic Structured Output Parsers, Redis Semantic Caching

Databases & Messaging: MongoDB, Redis (Geospatial & Caching), PostgreSQL, Mongoose, Apache Kafka (Event Streaming)

Cloud & DevOps: AWS (ECS Fargate, Lambda, S3, RDS, CloudWatch), Terraform (IaC), Docker, GitHub Actions CI/CD, Postman, Linux

PROFESSIONAL EXPERIENCE

Associate Engineer | Virtusa (India) Dec 2025 – Present

- Architect and maintain high-performance full-stack web applications using React 18, Next.js, Node.js, and TypeScript across enterprise engagements.
- Develop scalable RESTful microservice APIs and WebSocket endpoints with Express.js and MongoDB/PostgreSQL, optimizing query execution plans.
- Engineer intelligent workflow automations using LangChain LCEL and OpenAI APIs, implementing dynamic prompt chaining and structured output parsing.
- Collaborate in daily Agile Scrum sprints, code reviews, and containerized Docker deployments with GitHub Actions CI/CD.

KEY PROJECTS

Generative AI & LangChain Multi-Agent Architecture Toolkit | Python, LangChain, LangGraph, LCEL, OpenAI GPT-4o, ChromaDB, FastAPI, Docker GitHub

- Architected enterprise Generative AI toolkit featuring cyclical multi-agent orchestration via LangGraph, dynamic tool routing, and LangChain Expression Language (LCEL) Runnables (RunnableSequence, RunnableParallel, RunnablePassthrough).
- Engineered end-to-end RAG pipeline utilizing ChromaDB vector store, semantic document chunking, metadata filtering, and context-aware conversational memory with guaranteed JSON schema validation via Pydantic output parsers.
- Served low-latency streaming inference endpoints through Dockerized FastAPI microservice with real-time token delivery over WebSockets/SSE.

Oxima – Real-Time Fleet & Ride-Hailing Platform | React 18, Node.js, Express, Redis Geospatial, Apache Kafka, WebSockets, Tailwind GitHub

- Engineered scalable ride-hailing architecture with live GPS tracking, OpenStreetMap OSRM road routing, and Zustand reactive state management.
- Implemented sub-50ms driver-rider proximity matchmaking using Redis Geospatial indexes (GEOADD, GEORADIUS).
- Decoupled trip requests via Apache Kafka partitioned event queues and broadcasted telemetry over bidirectional WebSockets.

CampusConnect – Real-Time Student Marketplace | React, Node.js, Express, MongoDB, Socket.IO, Razorpay, Cloudinary GitHub

- Built campus marketplace enabling peer commerce, live chat rooms, and real-time notification feeds via Socket.IO.
- Implemented stateless JWT authentication, Google OAuth 2.0, Razorpay payment gateway integration, and Cloudinary media optimization.

EDUCATION & CERTIFICATIONS

Bachelor of Technology (B.Tech) in Computer Science & Engineering | DIT University, Dehradun 2021 – 2025

Certifications: AWS Certified Developer – Associate (DVA-C02) • AWS Certified Data Engineer – Associate (DEA-C01) • Node.js & Express (Coursera) • React Frontend Development (freeCodeCamp) • Agile Project Management (LinkedIn Learning)